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# DESIGN AND CONFIGURATION OF A SIMPLE INTERNETWORK USING A ROUTER

AIM:

To design and configure a simple internetwork using a router in Cisco Packet Tracer, assign static IP addresses to computers and router interfaces, and study packet transmission between different networks.

ALGORITHM:

1. Open Cisco Packet Tracer and create a new network topology.
2. Place a router, two switches, and four PCs in the workspace.
3. Connect two PCs to each switch using Copper Straight-Through cables.
4. Connect both switches to the router using Ethernet cables.
5. Assign static IP addresses and subnet masks to all PCs.
6. Configure the router interfaces with suitable IP addresses for both networks.
7. Enable the router interfaces using the no shutdown command.
8. Configure the default gateway on each PC using the IP address of its corresponding router interface.
9. Check connectivity between PCs in the same network using the ping command.
10. Send packets between PCs in different networks and observe the communication.
11. Use Simulation mode to trace packet transmission through the router.
12. Record the results and verify successful communication between the networks.

IP ADDRESS CONFIGURATION:

Network 1:

PC0 IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

PC1 IP Address: 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Router Interface G0/0: 192.168.1.1

Network 2:

PC2 IP Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

PC3 IP Address: 192.168.2.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

Router Interface G0/1: 192.168.2.1

ROUTER CONFIGURATION:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
end
```

COMMAND:

ping 192.168.2.2

OUTPUT:

Case 1: Successful Communication

When the router interfaces and IP addresses are configured correctly, the source PC can communicate with the destination PC in the other network.

Sample output:

Reply from 192.168.2.2: bytes=32 time=2ms TTL=127

Reply from 192.168.2.2: bytes=32 time=1ms TTL=127

Case 2: Packet Transmission in Simulation Mode

The packets travel from the source PC through the switch to the router. The router forwards them to the destination network through the appropriate interface.

RESULT:

Thus, a simple internetwork was designed and configured using Cisco Packet Tracer. Static IP addresses were assigned to the computers and router interfaces, and communication between different networks was tested successfully using the ping command.